

CFD analysis of a classic car - technical synthesis

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1. Case Study

- The analysis focuses on the external aerodynamics of a classic rally car running at constant speed equal to 30 m/s, with the goal of studying the steady-state flow behavior around its body and identifying the primary sources of aerodynamic drag and flow separation.
- The main objective of the simulation is to validate the numerical results against reference data, particularly the aerodynamic drag coefficient C_x , in order to evaluate the reliability of the CFD setup and support further aerodynamic studies on vintage motorsport vehicles.

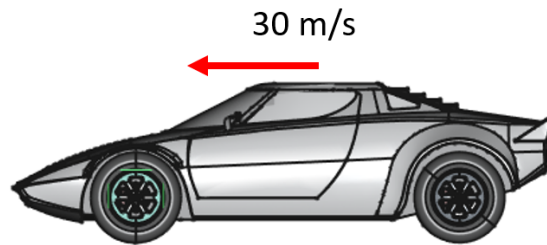


Figure 1 – Case scenario

2. CAD preparation

- The CAD geometry was downloaded from GrabCAD at the following [link](#).
- A cleanup process was carried out to remove unnecessary components such as internal parts, logos, screws, and hidden details that are irrelevant to the aerodynamic analysis.
- The model was then exported to STL format, separating the car body and wheels into different files to allow targeted meshing and boundary condition definitions.

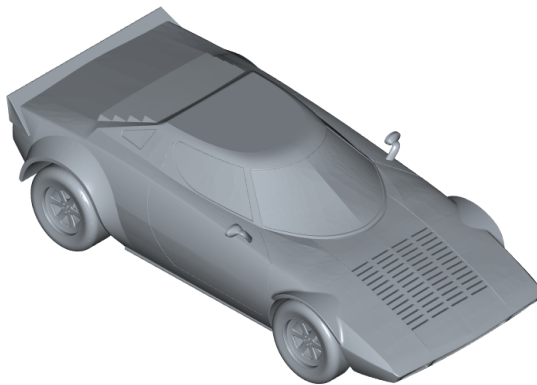


Figure 2 – STL

3. Mesh creation

- The mesh was generated through a two-step process. First, a background mesh was created using the blockMesh utility to define the computational domain. This domain was sized to be large enough

to not affect the aerodynamic phenomena near the vehicle, while avoiding excessive size that would unnecessarily increase the computational load.

- Subsequently, the `snappyHexMesh` tool was employed to generate the mesh around the STL geometry. Mesh refinement was applied in the region surrounding the vehicle, with additional refinement zones defined around the wheels.
- The final mesh consists of approximately 1.3 million cells, offering a balanced trade-off between computational cost and simulation accuracy for external aerodynamic analysis.

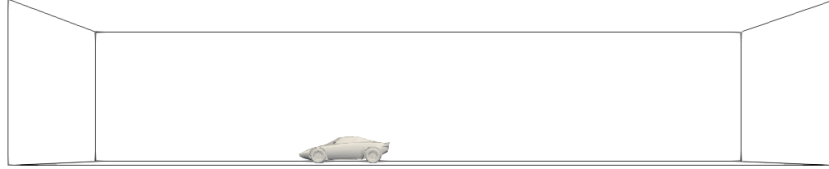


Figure 2 – Overview of the computational mesh

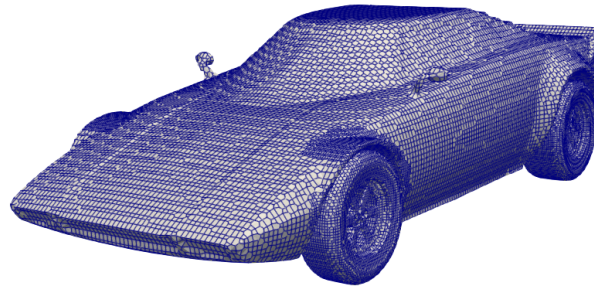


Figure 3 – Mesh detail around the vehicle body

4. Boundary Conditions Overview

- To simulate the external aerodynamics of a vehicle in a realistic and computationally efficient way, appropriate boundary conditions were applied to the airflow, pressure, and turbulence fields.
- A uniform incoming airflow was set to replicate typical driving conditions, while the outlet allowed the air to exit freely, ensuring numerical stability. The ground was modeled as a moving surface to reproduce the relative motion between the vehicle and the road, an approach commonly used in aerodynamic analyses. The top and lateral boundaries were treated as frictionless surfaces to avoid artificial confinement of the flow. The vehicle body was defined as a fixed wall, while the wheels were set to rotate realistically to capture their influence on the surrounding flow.
- Turbulence was modeled using standard industry approaches, with estimated inlet conditions and dedicated near-wall treatments on solid surfaces. These functions allow accurate prediction of aerodynamic forces such as drag and lift, while keeping computational demands manageable.

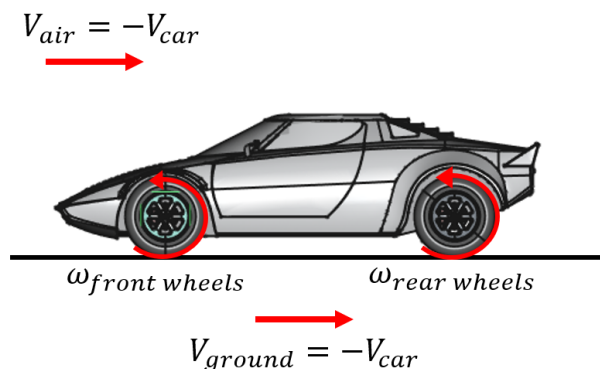


Figure 4 – Boundary conditions

5. Simulation Setup

- The simulation was carried out using OpenFOAM 12 within the foamRun application. The solver adopted is `incompressibleFluid`, which is suited for steady-state simulations involving incompressible turbulent flows. This approach enables the assessment of aerodynamic performance under constant driving conditions.
- The simulation runs from an initial time of zero up to a final time chosen to ensure convergence, with results saved periodically for post-processing. The time step was selected to ensure numerical stability and efficient progression of the simulation.

6. Results

- **Velocity Field.** The velocity field around the vehicle was analyzed to understand how the airflow interacts with its surfaces. The streamlines, which represent the path followed by fluid particles, were visualized to highlight regions of flow acceleration, separation, and recirculation. These visualizations help identify how the shape of the vehicle affects the surrounding flow and contributes to aerodynamic drag.

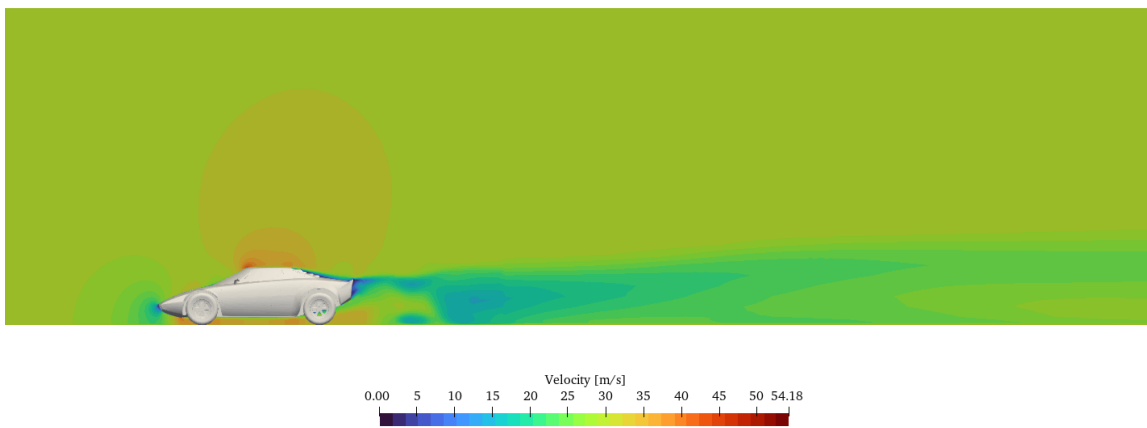


Figure 5 – Velocity Field

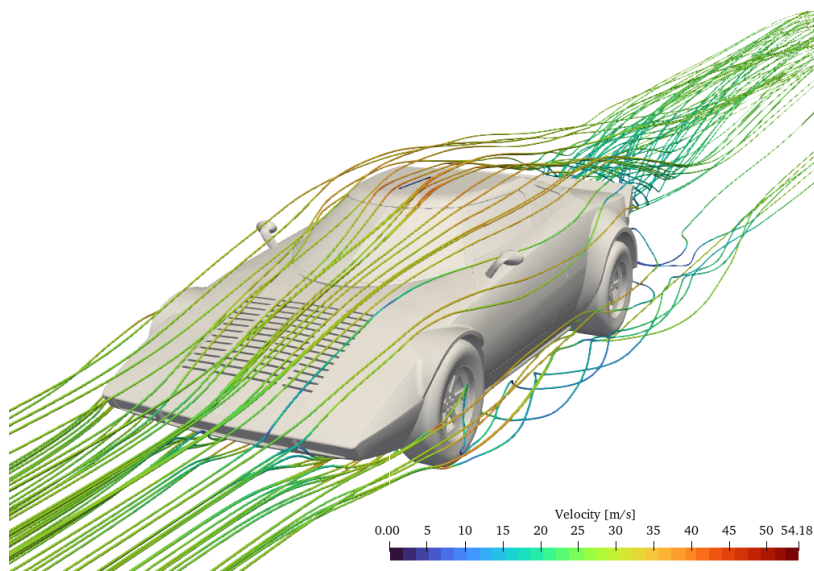


Figure 6 – Streamlines

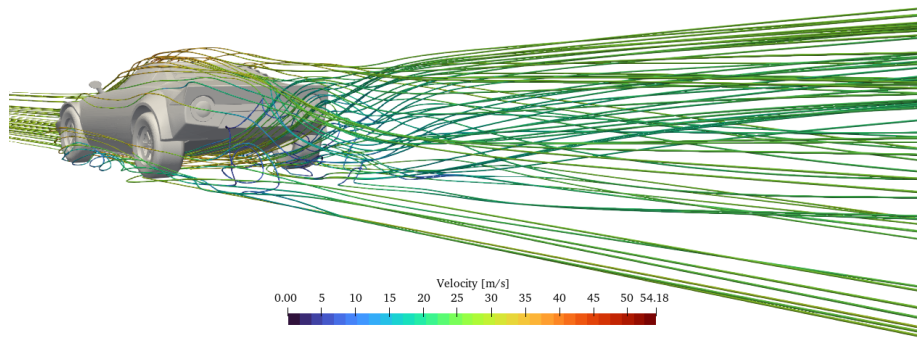


Figure 7 – Streamlines

- Pressure Coefficient.** The pressure distribution around the vehicle was evaluated through the pressure coefficient (C_p), a dimensionless quantity that highlights how pressure varies along the surfaces of the car. This coefficient acts along the surface normal and indicates where aerodynamic forces are generated and it is an indicator of where the drag and the lift are generated or recovered.

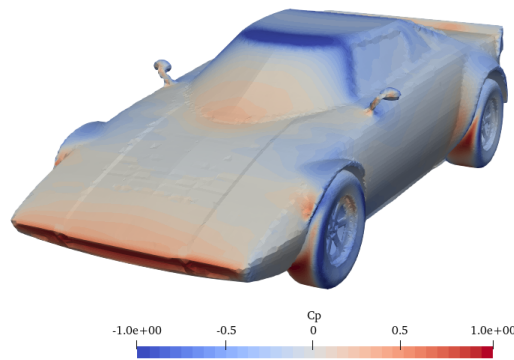


Figure 8 – Pressure Coefficient - Front view

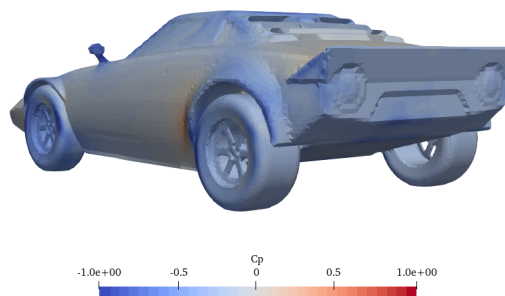


Figure 9 – Pressure Coefficient - Rear view

- Wake region analysis** The wake region behind the vehicle was analyzed using the Coefficient of Total Pressure (C_{ptot}), which provides insight into energy losses in the flow due to drag and turbulence. An isosurface at $C_{ptot} = 0$ was generated to visualize the boundary of the low-energy region behind the car. This surface highlights areas where the total pressure equals the freestream value, effectively enclosing the wake bubble. The shape and extent of this isosurface help assess the vehicle's aerodynamic efficiency and flow recovery downstream.

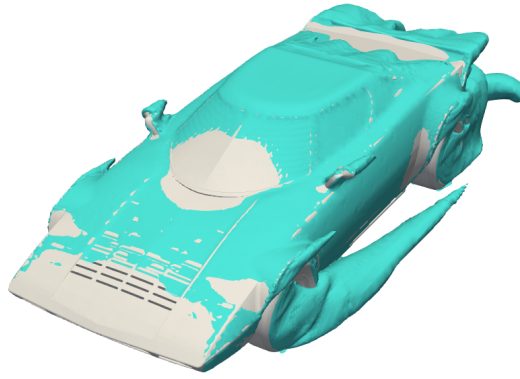


Figure 10 – Wake Region

- **Drag Coefficient** The drag coefficient was computed using an external function evaluated at each simulation step. The plot below shows its evolution over time, highlighting a progressive stabilization around the experimental reference value of 0.4.

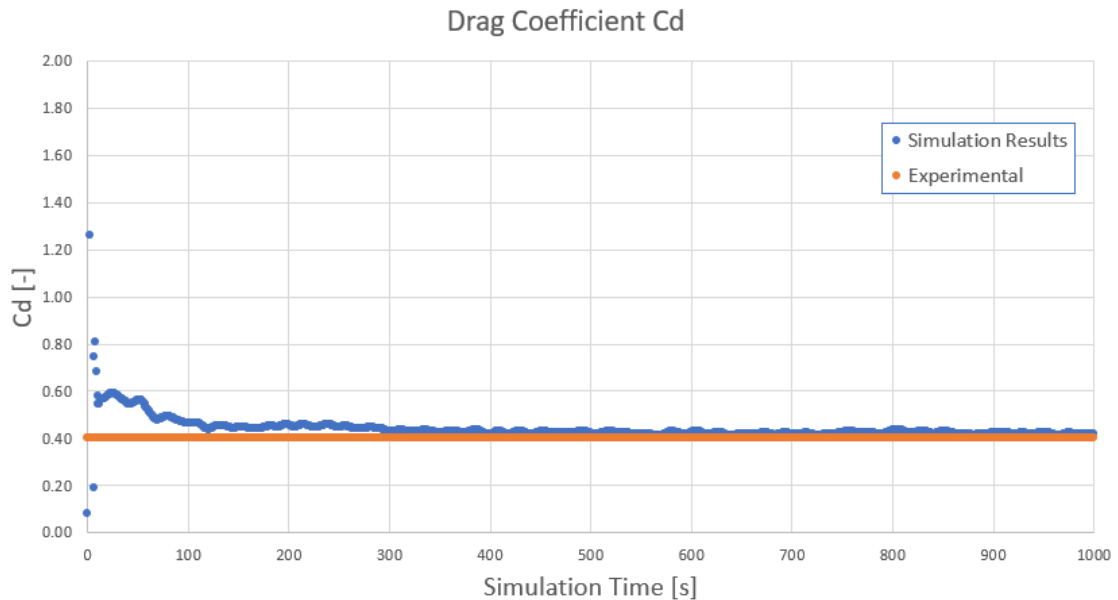


Figure 11 – Drag Coefficient

7. Comments and Next Steps

- The CFD analysis provides a general yet insightful overview of the vehicle's aerodynamic behavior, successfully capturing key flow features such as separation zones and wake structure.
- Despite the relatively low mesh resolution (approximately 1.3 million cells), the computed drag coefficient matches the known experimental value, supporting the validity of the simulation setup and modeling choices.
- As a next step, it would be beneficial to refine the mesh by increasing resolution in critical regions such as the side mirrors and rear spoiler. This would allow for more accurate force predictions and local flow visualization. Additionally, the model could be used to explore potential aerodynamic optimizations for the vehicle.

Genoa, August 5, 2025